JAHIN TAZWAR

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SUMMARY

Computer Science undergraduate at CUET focused on systems programming and developer tools. Independently built and deployed four end-to-end projects, including a programming language interpreter in C and a multithreaded chess engine in C++, both compiled to WebAssembly and running in the browser. Strong C/C++ and algorithms foundation with full-stack delivery experience.

TECHNICAL SKILLS

Languages: C, C++ (C++17), JavaScript, SQL
Web & Data: Node.js, Express.js, REST APIs, React, HTML/CSS, PostgreSQL, MongoDB, SQLite
Tools: Git, GitHub, CMake, WebAssembly (Emscripten), Linux, VS Code, Claude Code
Concepts: Data Structures & Algorithms, Compiler/Interpreter Design, OOP, Multithreading, Memory Management, Performance Optimization

PROJECTS

NFA — Programming Language & Interpreter nfa-lang.netlify.app
C, WebAssembly (Emscripten), JavaScript

- Built a complete interpreted language from scratch in C — hand-written lexer, recursive-descent parser, AST construction, tree-walking evaluator, and a block-scoped symbol table.
- Implemented variables, arrays, strings, numeric types, loops, and a built-in function library; shipped both a native CLI runner and an interactive REPL from one shared runtime core.
- Compiled the C runtime to WebAssembly so the language executes entirely client-side, with no backend infrastructure or server cost; managed manual allocation and object lifetimes leak-free across the execution loop.

NFA's Gambit — Chess Engine nfa-gambit.netlify.app
C++17, Multithreading, WebAssembly, CMake

- Engineered a full chess engine with legal move generation, board representation, positional evaluation, and a graphical interface.
- Implemented Minimax with Alpha-Beta pruning and Iterative Deepening, cutting nodes evaluated per depth by orders of magnitude versus naive minimax and enabling deeper lookahead in the same time budget.
- Applied modern C++ throughout — RAII and smart pointers for zero manual deallocation, an OO class hierarchy for pieces and game state, and multithreaded search; ported to WebAssembly for in-browser play.

Repo Explainer — AI Codebase Analysis Tool repodecode.netlify.app
JavaScript, React, Node.js, LLM APIs

- Built an AI tool that ingests any public GitHub repository and generates structured explanations of its architecture, module layout, and functionality.
- Designed the analysis and prompt pipeline to produce consistent, developer-readable output across repositories of varying size and language, cutting codebase onboarding from hours of file-by-file reading to a single pass.

CUET Lab Simulations — Educational Web Platform cuet-lab.netlify.app
React, JavaScript, HTML/CSS

- Built and deployed a platform hosting virtual lab simulations and course resources, centralizing materials previously scattered across sources and making them available on demand to an entire cohort.

EDUCATION

Chittagong University of Engineering & Technology (CUET) — B.Sc. in Computer Science & Engineering *Expected 2030*
Government City College, Chittagong — Higher Secondary Certificate (HSC), GPA 5.00 / 5.00 *2025*
Chittagong Government High School — Secondary School Certificate (SSC), GPA 5.00 / 5.00 *2023*

ACHIEVEMENTS & LANGUAGES

- CodeChef competitive programming rating 1279; practices algorithmic problem solving regularly.
- Took four software projects from concept to publicly deployed product, spanning systems programming, compilers, AI tooling, and full-stack web.
- Languages: Bengali (Native), English (Professional Working Proficiency).